

Physics Lecture Notes

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Chapter 1

Introduction

1.1 Purpose of these notes

These notes were written for students of Computer Science who meet physics not as their main field of study, but as a part of their broader scientific and technical education. For this reason, the purpose of this script is not to reproduce a complete textbook of physics. It is not an encyclopedia, and it is not a catalogue of formulas. Its purpose is more modest, but also more important: to introduce selected parts of physics carefully enough that the reader can understand how physical reasoning works.

Physics is often presented to students as a collection of equations. In such a presentation, learning physics seems to mean remembering formulas and knowing when to substitute numbers into them. This is not the approach of these notes. A formula is not the beginning of understanding. A formula is a compressed statement about a physical model. To understand it, we must know what system is being described, what assumptions have been made, what quantities occur in the formula, what has been neglected, and in what range the formula can be used.

For example, an equation such as

$$F = ma$$

is not only a recipe for calculating force, mass, or acceleration. It expresses a relation between interaction and change of motion. It says something about the system, about the role of mass, about the meaning of acceleration, and about the way in which a simplified model of motion is being constructed. If the symbols are treated only as places where numbers should be inserted, then the physical content of the equation is lost.

One of the main goals of these notes is therefore to teach the reader how to read equations. Reading an equation does not mean pronouncing the symbols. It means understanding what the symbols represent, what kind of dependence is described, which quantities are variable and which are fixed, what happens when one quantity changes, what the limiting cases are, and whether the result makes physical sense. This skill is essential not only in physics, but also in mathematics, engineering, computer science, data analysis, simulation, modelling, and many other technical disciplines.

1.2 Mathematics as a tool for understanding

The reader should not expect these notes to avoid mathematics. Physics without mathematics would lose much of its power. At the same time, mathematics will be used here as a tool for understanding, not as a barrier. We will begin with mathematical preliminaries: functions, graphs, variables, vectors, derivatives, integrals, units, dimensions, and simple equations describing change. These topics will not be developed as an independent mathematics course. They will be introduced only to the extent needed for physical modelling.

The important point is that each mathematical object should carry meaning. A derivative is not only a formal operation; in kinematics it becomes velocity or acceleration, a rate of change of position or velocity. An integral is not only a method of finding an antiderivative; it can describe accumulation, displacement, work, or total effect. A vector is not only an ordered list of numbers; it represents a quantity with magnitude and direction. A graph is not decoration; it is information about behaviour. A differential equation is not only an equation with derivatives; it is often a compact way of saying how a system evolves.

1.3 From situation to model and back

The central method used throughout these notes can be summarized as a movement from a real or simplified situation to an idealized model, then to a mathematical description, then to calculation or simulation, and finally back to interpretation and checking.

We start from a situation in the real world or from a simplified physical scenario. Then we decide what is relevant and what can be neglected. We choose a system, variables, assumptions, and an idealized model. After that, we translate the model into mathematics. We calculate, analyse, or simulate. Finally, we return to physics: we interpret the result, check the units, consider limiting cases, and ask whether the answer makes sense.

This movement is not always linear. In real physical reasoning, we often move back and forth between meaning and mathematics. A calculation may suggest a physical interpretation. A physical idea may suggest a mathematical simplification. A unit check may reveal an error. A limiting case may show that the model is incomplete. A graph may help us understand an equation. A simulation may help us see what a formula predicts. For this reason, the notes will repeatedly return to the same pattern: from physical meaning to mathematical model, and from mathematical result back to physical meaning.

1.4 Why kinematics and dynamics matter

Special attention will be given to kinematics and dynamics. These are not only introductory chapters that must be completed before moving to more advanced topics. They are the backbone of the course. Kinematics introduces position, time, motion, velocity, acceleration, functions of time, and graphical descriptions of change. Dynamics introduces force, interaction, mass, inertia, Newton's laws, equations of motion, initial conditions, and prediction. In these chapters, the reader first encounters the full structure of physical modelling.

Later topics, such as oscillations, waves, energy, fields, or selected ideas from modern physics, should not be seen as disconnected areas. They reuse the same habits of thought. Again and again, we ask: What is the system? What are the relevant quantities? What assumptions are we making? What mathematical relation describes the model? What can be predicted? What does the result mean?

1.5 Depth rather than a catalogue of formulas

The notes deliberately prefer depth over the number of formulas. It is better to understand a small number of fundamental ideas well than to collect many equations without knowing how to use them. For each important topic, we will try to identify the physical situation, build the simplest useful model, state the assumptions, derive or motivate the mathematical description, and interpret the result. Some computations will be simple on purpose. Their role is not to test algebraic endurance, but to reveal structure.

This is especially important because many standard textbooks leave gaps in the reasoning. They often move quickly from a physical description to a formula, or from a formula to a result, assuming that the reader can reconstruct the intermediate steps. These notes should be more explicit. They should show not only what is done, but why it is done. They should make visible the transition from words to equations, from equations to consequences, and from consequences back to interpretation.

1.6 Units, dimensions, and checks

Units and dimensions will play an important role in this process. Checking units is one of the simplest and most powerful ways to test whether a formula or result can be correct. If a calculation gives a time measured in kilograms, something has gone wrong. Dimensional analysis is not a cosmetic detail; it is a basic tool of independent thinking.

Throughout the notes, the reader should learn to ask: What unit should the answer have? Do both sides of the equation have the same dimension? Does the result become larger or smaller in the expected way? What happens in extreme cases? Does the answer make physical sense?

1.7 Approximation and idealization

The notes will also emphasize the role of approximation and idealization. Physics does not describe reality by copying it in all its complexity. Instead, it builds models. We may neglect air resistance, treat a body as a point mass, assume a frictionless surface, use a small-angle approximation, or describe a complicated interaction with a simple force law. Such simplifications are not mistakes if they are made consciously. A model is useful when it captures the relevant structure of a situation and allows us to reason, calculate, predict, and understand. But every model has limits, and those limits must be remembered.

1.8 Computational experiments and simulations

The course will not rely on traditional laboratory experiments. However, experimental thinking will still be present. In many places, the reader may work with computational experiments or simulations. For Computer Science students this is natural and useful. A simple program or an AI-assisted HTML application can simulate a pendulum, projectile motion, oscillations, or waves. Such a simulation can allow us to change parameters, collect artificial measurements, estimate periods or frequencies, compare results with theory, and discuss measurement uncertainty.

A simulation is not reality. It is also a model. But precisely because it is a model, it can make the structure of physical reasoning visible. It allows us to see how assumptions become equations, how equations produce motion, how measurements are extracted, and how predictions are checked. Used correctly, computational experiments can help connect physics, mathematics, programming, and interpretation.

1.9 How to study these notes

The reader should approach these notes actively. Do not only look for final formulas. Ask what problem is being solved. Ask what assumptions are being made. Ask why a given quantity was chosen. Ask what the equation says before using it. Ask whether the answer has the right unit. Ask what would change if the assumptions changed. Ask whether the result is physically reasonable.

The goal of the course is not only to pass an exam. Passing the course is important, but it is not the deepest purpose of the notes. The deeper goal is to develop a way of thinking: to see physics as a disciplined movement between reality, idealization, mathematics, computation, and interpretation. A student who follows this path should become more comfortable with formulas, more careful with assumptions, more aware of the meaning of mathematical models, and more capable of using quantitative reasoning in future technical subjects.

In this sense, physics is not merely another subject to memorize. It is a training ground for modelling. It teaches how to take a complex situation, simplify it without losing what is

essential, describe it mathematically, analyse the description, and return to the world with a prediction or explanation. This ability is valuable far beyond physics itself. It is one of the foundations of scientific and technical thinking.

Chapter 2

Mathematical Foundations

2.1 Why this chapter is necessary

This chapter collects the minimal mathematical language that will be used throughout these notes. It is not a replacement for a mathematics course. The reader is expected to have already met elementary algebra, functions, analytic geometry, vectors, matrices, derivatives, and integrals. However, experience shows that having passed a mathematics course does not always mean being able to use mathematics fluently in a new context.

The purpose of this chapter is therefore practical. We recall the mathematical objects and notations that will appear later, before they are used in physical models. At this stage we do not develop the physical applications in detail. Those applications will appear in later chapters, especially in kinematics and dynamics. Here the goal is to make the mathematical language readable.

Mathematics in these notes should not be treated as a collection of symbolic tricks. A formula is not only something into which numbers are substituted. A formula expresses a relation. A function describes dependence. A derivative describes local change. An integral describes accumulation. A vector describes magnitude and direction. A matrix can describe a transformation or organize several quantities at once.

The reader should use this chapter as a short toolbox. The tools introduced here will return later with physical meaning.

2.2 Scientific notation

Physics and science in general often deal with very large and very small numbers. Writing such numbers in ordinary decimal notation is inconvenient and often makes the scale of the quantity difficult to see. For this reason we use scientific notation.

A number written in scientific notation has the form

$$a \times 10^n,$$

where n is an integer and usually

$$1 \leq |a| < 10.$$

The number a tells us the significant digits, while 10^n tells us the scale.

For example,

$$2.7 \times 10^{17}$$

means

$$270000000000000000.$$

The exponent 17 tells us that the decimal point has been moved seventeen places to the right.

Similarly,

$$1.602 \times 10^{-19}$$

means

$$0.0000000000000000001602.$$

The exponent -19 tells us that the decimal point has been moved nineteen places to the left.

This notation is not optional. It is a basic part of scientific language. Without it, many numbers used in science would be almost unreadable.

For example, the elementary charge is approximately

$$e = 1.602 \times 10^{-19} \text{ C}.$$

Written without scientific notation, this is

$$e = 0.0000000000000000001602 \text{ C}.$$

The scientific form is much easier to read and compare.

The Avogadro constant is approximately

$$N_A = 6.022 \times 10^{23} \text{ mol}^{-1}.$$

Written in full decimal form, this is approximately

$$N_A = 602200000000000000000000 \text{ mol}^{-1}.$$

The scientific notation immediately shows that this is a number of order 10^{23} .

The speed of light in vacuum is approximately

$$c = 3.00 \times 10^8 \text{ m/s}.$$

In ordinary decimal notation, this is

$$c = 300000000 \text{ m/s}.$$

Scientific notation also makes multiplication and division easier. We use the rules

$$10^a \cdot 10^b = 10^{a+b}$$

and

$$\frac{10^a}{10^b} = 10^{a-b}.$$

For example,

$$(2 \times 10^3)(3 \times 10^4) = 6 \times 10^{3+4} = 6 \times 10^7.$$

Similarly,

$$\frac{8 \times 10^6}{2 \times 10^2} = 4 \times 10^{6-2} = 4 \times 10^4.$$

The exponent of ten gives the order of magnitude. In many scientific situations, understanding the order of magnitude is more important than knowing many decimal digits. A result of order 10^{-19} and a result of order 10^{19} describe completely different scales.

The reader is expected to understand notation such as

$$10^{-6}, \quad 10^3, \quad 2.7 \times 10^{17}, \quad 6.022 \times 10^{23}.$$

Without this basic skill, it is not possible to read many scientific formulas or numerical examples.

2.3 Numbers, quantities, and units

A number by itself is often not enough. In science, we usually work with quantities. A quantity has a numerical value, a unit, and a meaning.

For example, the expression

$$5$$

is only a number. It becomes a physical quantity only when we know what it represents:

$$5 \text{ m}, \quad 5 \text{ s}, \quad 5 \text{ kg}, \quad 5 \text{ C}.$$

These expressions do not mean the same thing. The unit is part of the information.

Units must be treated consistently. We can add quantities with the same unit:

$$2 \text{ m} + 3 \text{ m} = 5 \text{ m}.$$

But the expression

$$2 \text{ m} + 3 \text{ s}$$

is not a meaningful sum, because metres and seconds describe different kinds of quantities.

Units also help us check formulas. If the left-hand side of an equation has a certain unit, then the right-hand side must have the same unit. This simple observation is one of the most useful ways to detect errors.

For example, if a formula gives a quantity measured in metres, every term that is added to produce that quantity must also have the unit of metres. If a calculation produces seconds where metres are expected, something is wrong.

This type of check does not solve the problem for us, but it protects us from many meaningless results.

2.4 Variables, constants, and parameters

Mathematical notation uses symbols. These symbols may play different roles. Some symbols represent variables, some represent constants, and some represent parameters.

A variable is a quantity that is allowed to change. For example, in the expression

$$f(x) = x^2 + 1,$$

the symbol x is the variable. Different values of x give different values of $f(x)$.

A constant has a fixed value. For example, in

$$f(x) = 3x + 2,$$

the numbers 3 and 2 are constants.

A parameter is a symbol that is treated as fixed in one situation, but may have different values in different situations. For example,

$$f(x) = ax + b$$

describes a whole family of linear functions. For a particular function, a and b are fixed. But if we choose different values of a and b , we get a different function.

For example, if

$$a = 2, \quad b = 5,$$

then

$$f(x) = 2x + 5.$$

If

$$a = -1, \quad b = 0,$$

then

$$f(x) = -x.$$

When reading a formula, we should ask:

- Which symbols are variables?
- Which symbols are constants?
- Which symbols are parameters?
- What values are allowed?
- What changes when one symbol changes?
- What remains fixed?

This distinction is essential for understanding mathematical models.

2.5 Functions

A function assigns an output to an input. If f is a function and x is an input, then

$$f(x)$$

denotes the output corresponding to x .

For example, let

$$f(x) = x^2 + 1.$$

Then

$$f(0) = 0^2 + 1 = 1,$$

$$f(1) = 1^2 + 1 = 2,$$

and

$$f(3) = 3^2 + 1 = 10.$$

The notation $f(x)$ should not be read as multiplication of f and x . It means: the value of the function f at the argument x .

A function may also depend on more than one variable. For example,

$$g(x, y) = x^2 + y^2$$

has two inputs. If

$$x = 3, \quad y = 4,$$

then

$$g(3, 4) = 3^2 + 4^2 = 25.$$

Functions are important because they describe dependence. When we write

$$y = f(x),$$

we say that y depends on x . The question is not only how to compute values, but also how to understand the dependence.

For example, if

$$f(x) = 2x + 1,$$

then increasing x by 1 increases $f(x)$ by 2. If

$$f(x) = x^2,$$

then the change in $f(x)$ depends on the current value of x . The function is no longer linear.

2.6 Cartesian coordinates

To describe points in a line, plane, or space, we use coordinates. The most common coordinate system is the Cartesian coordinate system.

On a line, a point is described by one coordinate:

$$x.$$

In a plane, a point is described by two coordinates:

$$(x, y).$$

In space, a point is described by three coordinates:

$$(x, y, z).$$

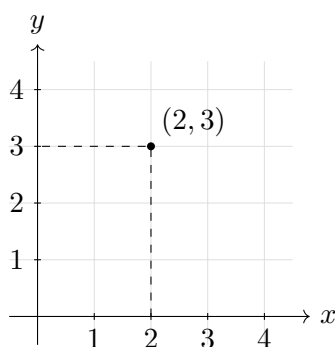
For example, the point

$$(2, 3)$$

in the plane has first coordinate 2 and second coordinate 3. The point

$$(1, -2, 5)$$

in space has coordinates 1, -2 , and 5.



This is naturally connected with the way we describe the world around us. We live in space, and many situations require us to say where something is. A coordinate system gives a mathematical way of doing this.

However, coordinates are not the point itself. Coordinates depend on a choice of origin, axes, orientation, and units. The same point can have different coordinates in different coordinate systems.

This distinction is important. Geometry concerns objects such as points, lines, planes, and curves. Coordinates are a method of describing these objects using numbers.

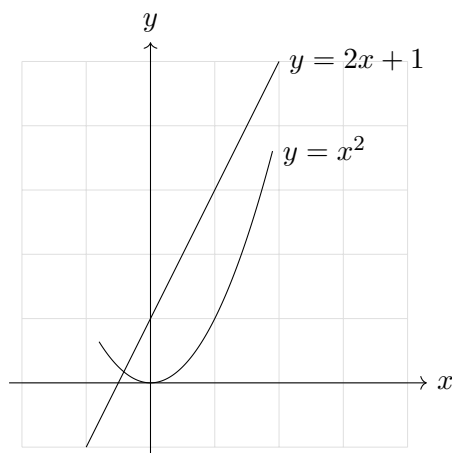
2.7 Graphs

A graph is a visual representation of a relation. For a function

$$y = f(x),$$

we usually place x on the horizontal axis and y on the vertical axis. Each point on the graph has coordinates

$$(x, f(x)).$$



A graph is not only a picture. It contains information. From a graph we can often see:

- where the function is increasing,
- where it is decreasing,
- where it is constant,
- where it changes rapidly,
- where it changes slowly,
- whether it is linear or nonlinear,
- whether it has zeros,
- whether it has maximum or minimum values.

For example, the graph of

$$f(x) = 2x + 1$$

is a straight line. The graph of

$$f(x) = x^2$$

is a parabola.

The reader should become comfortable moving between formulas and graphs. A formula gives a symbolic description. A graph gives a visual description. They are two ways of representing the same relation.

2.8 Curves and parametric descriptions

A curve can be described in different ways. One way is to give an equation such as

$$y = f(x).$$

For example,

$$y = x^2$$

describes a parabola in the plane.

Another way is to use a parameter. A parameter is an additional variable that describes how the coordinates are generated. For example, we can write

$$x = t, \quad y = t^2.$$

As t changes, the point

$$(x, y) = (t, t^2)$$

moves along the parabola.

A parametric curve in the plane can be written as

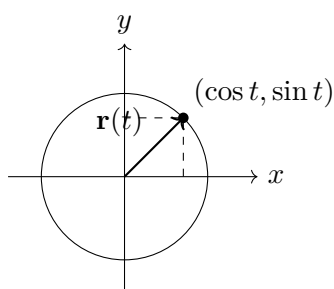
$$\mathbf{r}(t) = (x(t), y(t)).$$

For example,

$$\mathbf{r}(t) = (\cos t, \sin t)$$

describes the unit circle, because

$$\cos^2 t + \sin^2 t = 1.$$



At this stage, the parameter t is only a mathematical parameter. In later chapters it may be interpreted in a physical way, but here we only need the mathematical idea: a curve can be described by giving its coordinates as functions of a parameter.

2.9 Vectors

A vector is a mathematical object with magnitude and direction. In coordinates, a vector in the plane may be written as

$$\mathbf{v} = (v_x, v_y),$$

and a vector in space as

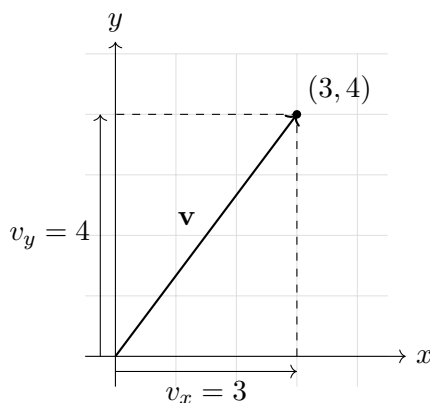
$$\mathbf{v} = (v_x, v_y, v_z).$$

The numbers v_x , v_y , and v_z are called components. They depend on the coordinate system. For example,

$$\mathbf{v} = (3, 4)$$

has magnitude

$$|\mathbf{v}| = \sqrt{3^2 + 4^2} = 5.$$



Vectors can be added component by component:

$$(1, 2) + (3, 4) = (4, 6).$$

They can be multiplied by a scalar:

$$2(1, 3) = (2, 6).$$

The zero vector is

$$\mathbf{0} = (0, 0)$$

in the plane, or

$$\mathbf{0} = (0, 0, 0)$$

in space.

The dot product of two vectors is defined by

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y$$

in two dimensions, and by

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z$$

in three dimensions.

The dot product is also related to the angle between vectors:

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta.$$

This formula shows that the dot product contains information about both lengths and direction.

Vectors will be used later whenever direction matters. Here the important point is only the mathematical structure: components, magnitude, addition, scalar multiplication, and dot product.

2.10 Matrices

A matrix is a rectangular array of numbers. For example,

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

Matrices can be added when they have the same size:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 10 & 12 \end{pmatrix}.$$

A matrix can be multiplied by a scalar:

$$2 \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix}.$$

A matrix can also multiply a vector. For example,

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + 2y \\ 3x + 4y \end{pmatrix}.$$

Matrix multiplication is not the same as ordinary multiplication of numbers. In general,

$$AB \neq BA.$$

This means that the order of multiplication matters.

Matrices are useful because they can represent linear transformations. For example, the matrix

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

represents a rotation in the plane by an angle θ . If

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix},$$

then

$$R(\theta)\mathbf{x}$$

is the rotated vector.

In these notes, matrices will not be developed in detail. The reader should remember only the basic idea: matrices organize several numbers and can describe transformations acting on vectors or systems of quantities.

2.11 Derivatives

The derivative describes how a function changes locally. If

$$y = f(x),$$

then the derivative of f with respect to x is written as

$$f'(x)$$

or

$$\frac{df}{dx}.$$

Intuitively, the derivative tells us how much the value of the function changes when the input changes by a very small amount. It is the local rate of change of the function.

For example, if

$$f(x) = x^2,$$

then

$$f'(x) = 2x.$$

At $x = 1$,

$$f'(1) = 2.$$

At $x = 3$,

$$f'(3) = 6.$$

This means that the function x^2 changes more rapidly near $x = 3$ than near $x = 1$.

If

$$f(x) = 3x + 2,$$

then

$$f'(x) = 3.$$

The derivative is constant because the function is linear and has the same slope everywhere.

If

$$f(x) = C,$$

where C is a constant, then

$$f'(x) = 0.$$

A constant function does not change.

The derivative also has a geometric interpretation. It is the slope of the tangent line to the graph of the function. If the derivative is positive, the function is increasing. If the derivative is negative, the function is decreasing. If the derivative is zero, the function is locally flat.

Some basic differentiation rules are:

$$\frac{d}{dx}x^n = nx^{n-1},$$

$$\frac{d}{dx}(f(x) + g(x)) = f'(x) + g'(x),$$

$$\frac{d}{dx}(cf(x)) = cf'(x),$$

where c is a constant.

For example,

$$\frac{d}{dx}(3x^2 + 5x - 7) = 6x + 5.$$

In later chapters, derivatives will receive physical interpretations. In this chapter, the important point is purely mathematical: a derivative measures local change.

2.12 Integrals

An integral describes accumulation. It can be understood as a limiting sum or, geometrically, as the signed area under a graph.

The definite integral of $f(x)$ from a to b is written as

$$\int_a^b f(x) dx.$$

For example,

$$\int_0^3 2x dx = [x^2]_0^3 = 9 - 0 = 9.$$

The symbol dx indicates the variable of integration. In the expression

$$\int_a^b f(x) dx,$$

we integrate with respect to x .

The indefinite integral is written as

$$\int f(x) dx.$$

It gives a family of functions whose derivative is $f(x)$. For example,

$$\int 2x dx = x^2 + C,$$

because

$$\frac{d}{dx}(x^2 + C) = 2x.$$

The constant C appears because the derivative of a constant is zero. Therefore many functions can have the same derivative.

Some basic integration rules are:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

for $n \neq -1$,

$$\int (f(x) + g(x)) dx = \int f(x) dx + \int g(x) dx,$$

and

$$\int cf(x) dx = c \int f(x) dx,$$

where c is a constant.

For example,

$$\int (3x^2 + 5x - 7) dx = x^3 + \frac{5}{2}x^2 - 7x + C.$$

In later chapters, integrals will receive physical interpretations. Here the important mathematical idea is that integration describes accumulation and is closely connected with differentiation.

2.13 Differentiation and integration as inverse operations

Differentiation and integration are closely related. In a basic sense, they are inverse operations.

If

$$F'(x) = f(x),$$

then

$$\int f(x) dx = F(x) + C.$$

For example,

$$\frac{d}{dx}x^3 = 3x^2.$$

Therefore,

$$\int 3x^2 dx = x^3 + C.$$

For definite integrals, the connection is expressed by the fundamental theorem of calculus:

$$\int_a^b f(x) dx = F(b) - F(a),$$

where

$$F'(x) = f(x).$$

For example, if

$$f(x) = 2x,$$

then we can choose

$$F(x) = x^2.$$

Thus

$$\int_1^4 2x dx = F(4) - F(1) = 4^2 - 1^2 = 15.$$

This relation between local change and accumulation is one of the central ideas of calculus.

2.14 Simple differential equations

A differential equation is an equation involving an unknown function and one or more of its derivatives. It describes a relation between a function and its change.

For example,

$$\frac{dy}{dx} = 3$$

asks for a function whose derivative is 3. The general solution is

$$y(x) = 3x + C.$$

Another example is

$$\frac{dy}{dx} = 2x.$$

A solution is

$$y(x) = x^2 + C,$$

because

$$\frac{d}{dx}(x^2 + C) = 2x.$$

A slightly different example is

$$\frac{dy}{dx} = ky,$$

where k is a constant. This equation says that the derivative of the function is proportional to the function itself. One family of solutions is

$$y(x) = Ce^{kx}.$$

Indeed,

$$\frac{d}{dx}Ce^{kx} = kCe^{kx} = ky(x).$$

The constant C is determined by an additional condition. For example, if

$$y(0) = 2,$$

then

$$2 = Ce^{k \cdot 0} = C,$$

so

$$y(x) = 2e^{kx}.$$

In these notes, differential equations will appear later as mathematical descriptions of change. Here the important point is only the basic structure: a differential equation contains derivatives and asks for an unknown function.

2.15 Approximation and limiting cases

Mathematics often uses exact expressions, but in applications we frequently use approximations. An approximation replaces a complicated expression with a simpler one under specified conditions.

For example, for small x measured in radians,

$$\sin x \approx x.$$

Also, for small x ,

$$(1 + x)^n \approx 1 + nx.$$

Approximations are useful only when we know when they are valid. The expression

$$\sin x \approx x$$

is useful for small x , but not for large x .

A limiting case is a way of checking an expression by asking what happens when a variable becomes very small, very large, or takes a special value.

For example, consider

$$f(x) = \frac{x}{1+x}.$$

If x is very small, then

$$f(x) \approx x.$$

If x is very large, then

$$f(x) \approx 1.$$

These limiting behaviours help us understand the structure of the function.

In later chapters, approximations and limiting cases will help us check whether mathematical models make sense.

2.16 What the reader should remember

This chapter is only a mathematical preparation. It does not contain the physical theory itself. Its purpose is to make the later chapters readable.

The reader should remember the following points:

- Scientific notation is necessary for reading very large and very small numbers.
- A quantity has a numerical value, a unit, and a meaning.
- Variables, constants, and parameters play different roles in formulas.
- A function describes dependence.
- Cartesian coordinates describe points after a coordinate system has been chosen.
- A graph represents a relation visually.
- Parametric descriptions allow us to represent curves using an additional variable.
- Vectors have components, magnitude, and direction.
- Matrices organize numbers and can represent transformations.
- A derivative describes local change.
- An integral describes accumulation.
- Differentiation and integration are inverse operations in the sense of calculus.
- A differential equation is an equation for an unknown function involving derivatives.
- Approximations and limiting cases help us understand expressions and check results.

The physical meaning of these tools will be developed later. In particular, kinematics and dynamics will show how this mathematical language becomes a way of describing motion, interaction, and change.

Chapter 3

Kinematics

3.1 What kinematics studies

Kinematics is the part of mechanics that describes motion. At this stage we do not ask why the motion happens. We do not introduce forces, interactions, or causes of motion. Those questions belong to dynamics. In kinematics we ask a more basic question:

How can motion be described mathematically?

To answer this question, we need the mathematical language recalled in the previous chapter: coordinates, functions, parametric curves, vectors, derivatives, and integrals. The new idea is not a new mathematical object. The new idea is interpretation. We now use mathematical objects to describe the motion of a body in space and time.

The central object of kinematics is the position of a moving point as a function of time. Once position is known, we can ask how it changes. This leads to velocity. Once velocity is known, we can ask how velocity changes. This leads to acceleration. The structure is therefore

position $\longrightarrow$ velocity $\longrightarrow$ acceleration.

Mathematically, this structure is built using derivatives. Later we will also move in the opposite direction:

acceleration $\longrightarrow$ velocity $\longrightarrow$ position.

Mathematically, this opposite direction is built using integrals and initial conditions.

This chapter should be read slowly. The goal is not only to remember definitions. The goal is to understand how a mathematical description of motion is constructed.

3.2 Position as a function of time

3.2.1 Position in a coordinate system

To describe motion, we must first describe position. A position is described only after we choose a coordinate system. In one dimension, the position of a point can be represented by one coordinate:

$$x.$$

In two dimensions, the position is represented by two coordinates:

$$(x, y).$$

In three dimensions, the position is represented by three coordinates:

$$(x, y, z).$$

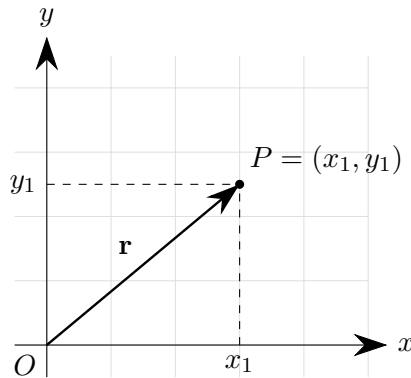
It is often convenient to write position as a vector. In three dimensions we write

$$\mathbf{r} = (x, y, z).$$

In two dimensions we write

$$\mathbf{r} = (x, y).$$

The symbol $\mathbf{r}$ represents the position vector. It points from the origin of the chosen coordinate system to the point whose position we want to describe.



The position vector depends on the coordinate system. If we choose a different origin or different axes, the coordinates of the same physical point may change. This is not a contradiction. Coordinates are not the point itself. Coordinates are a numerical description of the point after a coordinate system has been chosen.

3.2.2 Motion as changing position

A body is in motion when its position changes in time. This means that the coordinates of the point are functions of time.

In one dimension we write

$$x = x(t).$$

In two dimensions we write

$$x = x(t), \quad y = y(t),$$

or, in vector form,

$$\mathbf{r}(t) = (x(t), y(t)).$$

In three dimensions we write

$$\mathbf{r}(t) = (x(t), y(t), z(t)).$$

This notation is important. The symbol t represents time. The expression $\mathbf{r}(t)$ means that position depends on time. If we choose a particular value of time, we obtain the position of the body at that time.

For example, if

$$\mathbf{r}(t) = (t, t^2),$$

then at $t = 0$,

$$\mathbf{r}(0) = (0, 0).$$

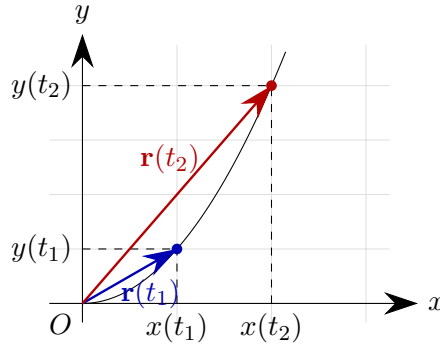
At $t = 1$,

$$\mathbf{r}(1) = (1, 1).$$

At $t = 2$,

$$\mathbf{r}(2) = (2, 4).$$

Thus one formula describes many positions. It tells us where the point is at each time.



The curve shows the possible positions of the point. The labels t_1 and t_2 mark two different moments of time. At each moment, the point has a different position vector. The vector itself is not the object that moves; it is the mathematical description of the position of the moving point at that time.

3.2.3 Trajectory

The set of all positions occupied by a moving point is called its trajectory. The trajectory is the curve traced by the moving point in space.

It is important to distinguish between the motion and the trajectory. The trajectory is the geometric curve. The motion contains more information, because it tells us where the point is at each moment of time.

Two different motions may have the same trajectory. For example,

$$\mathbf{r}_1(t) = (\cos t, \sin t)$$

and

$$\mathbf{r}_2(t) = (\cos 2t, \sin 2t)$$

both describe points moving on the unit circle, because in both cases

$$x^2 + y^2 = 1.$$

However, these two motions are not identical. The second parametrization goes around the same circle differently as time changes. The path is the same, but the time description is different.

This is one reason why parametric descriptions are so important in physics. Geometry alone tells us the shape of the path. Kinematics needs more: it needs the position as a function of time.

3.2.4 Motion along a straight line

The simplest example is motion along a straight line. In one dimension we may write

$$x(t) = x_0 + at,$$

where x_0 and a are constants. At this stage a is only a parameter in the formula. We will interpret such parameters more deeply after introducing velocity.

If $x_0 = 1$ and $a = 2$, then

$$x(t) = 1 + 2t.$$

At $t = 0$,

$$x(0) = 1.$$

At $t = 1$,

$$x(1) = 3.$$

At $t = 2$,

$$x(2) = 5.$$

The position changes along one axis. The trajectory is a line. The formula tells us not only the line on which the point lies, but also the position of the point at every time.

In two dimensions, a straight-line motion can be written as

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{a},$$

where

$$\mathbf{r}_0 = (x_0, y_0)$$

and

$$\mathbf{a} = (a_x, a_y).$$

In coordinates this means

$$x(t) = x_0 + a_x t,$$

$$y(t) = y_0 + a_y t.$$

As t changes, the point moves along a straight line in the plane.

3.2.5 Motion along a parabola

Consider the position

$$\mathbf{r}(t) = (t, t^2).$$

This means

$$x(t) = t,$$

and

$$y(t) = t^2.$$

If we eliminate the parameter t , we get

$$t = x,$$

and therefore

$$y = x^2.$$

So the trajectory is a parabola.
The parametric description

$$\mathbf{r}(t) = (t, t^2)$$

contains time information. The equation

$$y = x^2$$

contains only the shape of the trajectory. It tells us which curve is traced, but not how the point moves along the curve as time passes.

Another parametrization may give the same parabola with a different time description. For example,

$$\mathbf{r}(t) = (2t, 4t^2)$$

also satisfies

$$y = x^2,$$

because if $x = 2t$, then $t = x/2$, and

$$y = 4t^2 = 4\left(\frac{x}{2}\right)^2 = x^2.$$

The same geometric curve can therefore be produced by different time parametrizations.

3.2.6 Motion along a circle

Now consider

$$\mathbf{r}(t) = (R \cos t, R \sin t),$$

where $R > 0$ is a constant.

Then

$$x(t) = R \cos t,$$

and

$$y(t) = R \sin t.$$

We can check what geometric curve this describes. We compute

$$x^2(t) + y^2(t) = R^2 \cos^2 t + R^2 \sin^2 t.$$

Using

$$\cos^2 t + \sin^2 t = 1,$$

we obtain

$$x^2(t) + y^2(t) = R^2.$$

Thus the point moves along a circle of radius R centered at the origin.
Again, the equation

$$x^2 + y^2 = R^2$$

describes the circle as a geometric object. The parametric formula

$$\mathbf{r}(t) = (R \cos t, R \sin t)$$

describes the position on the circle at each time.

We may also write

$$\mathbf{r}(t) = (R \cos(\omega t), R \sin(\omega t)),$$

where ω is a constant. This describes the same circle, but the time dependence is different. The parameter ω controls how rapidly the argument of the trigonometric functions changes. We will return to the meaning of this when we discuss velocity and circular motion.

3.2.7 What position tells us and what it does not tell us

A function $\mathbf{r}(t)$ is the complete kinematic description of position. It tells us where the point is at every time. From it, we can find the trajectory by looking at the set of all points $\mathbf{r}(t)$.

However, position alone is not the end of the description. Once we know how position depends on time, we naturally ask how fast the position changes and in what direction it changes. This question leads to velocity.

Before moving on, the reader should remember the following points:

- Motion is described by position as a function of time.
- In one dimension, position is written as $x(t)$.
- In two or three dimensions, position is written as a vector $\mathbf{r}(t)$.
- The trajectory is the curve traced by the moving point.
- The trajectory is not the same as the full motion.
- Different time parametrizations may describe the same geometric curve.
- In physics, the time dependence is essential.

3.3 Velocity

3.3.1 Why position is not enough

Knowing the position of a moving point at every time gives a complete description of where the point is. But we often want to describe motion more locally. We want to know how the position is changing at a given moment.

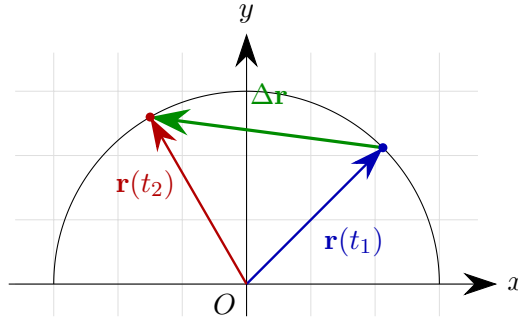
If the position changes a lot during a short time interval, we say that the motion is fast. If the position changes only a little during the same time interval, we say that the motion is slow. If the position changes in a different direction, the motion has a different direction.

Velocity is the mathematical object that describes this local change of position.

3.3.2 Average velocity

Let the position of a moving point be $\mathbf{r}(t)$. Suppose we compare two moments of time, t_1 and t_2 . The change in position is

$$\Delta \mathbf{r} = \mathbf{r}(t_2) - \mathbf{r}(t_1).$$



The vector $\Delta \mathbf{r}$ connects the position at time t_1 to the position at time t_2 . It is the difference between the two position vectors, not a separate vector floating independently of the motion.

The change in time is

$$\Delta t = t_2 - t_1.$$

The average velocity over this time interval is defined as

$$\mathbf{v}_{\text{avg}} = \frac{\Delta \mathbf{r}}{\Delta t} = \frac{\mathbf{r}(t_2) - \mathbf{r}(t_1)}{t_2 - t_1}.$$

This is a vector. Its direction is the direction of the displacement $\Delta \mathbf{r}$, and its magnitude tells us how much displacement occurs per unit time during the interval.

In one dimension, this becomes

$$v_{\text{avg}} = \frac{x(t_2) - x(t_1)}{t_2 - t_1}.$$

Average velocity depends on the interval. If we choose a large interval, we describe the overall change during that interval. If we choose a smaller interval, we get a more local description.

3.3.3 Instantaneous velocity

Average velocity describes motion over a time interval. To describe motion at one instant, we make that interval smaller and smaller.

Fix a time t . Instead of comparing two arbitrary times t_1 and t_2 , compare the position at time t with the position a short time Δt later:

$$\mathbf{r}(t + \Delta t) - \mathbf{r}(t).$$

The average velocity over this short interval is

$$\frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}.$$

This vector still describes an interval, not a single instant. To obtain the velocity at the instant t , we let the time interval shrink to zero:

$$\mathbf{v}(t) = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}.$$

This limit is the derivative of position with respect to time, so we write

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}.$$

The limiting process is the essential idea. We are not dividing by zero. We first compute the average velocity over a nonzero time interval Δt , and only then study what this expression approaches as Δt becomes arbitrarily small.

If

$$\mathbf{r}(t) = (x(t), y(t), z(t)),$$

then

$$\mathbf{v}(t) = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right).$$

In two dimensions,

$$\mathbf{v}(t) = \left(\frac{dx}{dt}, \frac{dy}{dt} \right).$$

In one dimension,

$$v(t) = \frac{dx}{dt}.$$

Velocity is therefore the derivative of position with respect to time. This is the first major connection between kinematics and calculus.

3.3.4 Velocity as a vector attached to the moving point

Velocity is a vector quantity. This means that it has magnitude and direction.

At each time t , the point has a position $\mathbf{r}(t)$ and a velocity $\mathbf{v}(t)$. It is useful to imagine the velocity vector as attached to the point at its current position.

The velocity vector is tangent to the trajectory. This statement is important. The trajectory tells us the path. The velocity tells us the local direction in which the point is moving along that path.

For example, if a point moves along a circle, the velocity is not directed toward the center of the circle. The velocity is tangent to the circle. The direction of velocity changes from point to point, even if the length of the velocity vector remains constant.

3.3.5 Speed

The magnitude of the velocity vector is called speed:

$$v(t) = |\mathbf{v}(t)|.$$

This notation can be slightly confusing because the same letter v is often used for both velocity and speed. In these notes, bold symbols such as $\mathbf{v}$ denote vectors, while non-bold symbols such as v may denote magnitudes.

If

$$\mathbf{v}(t) = (v_x(t), v_y(t), v_z(t)),$$

then

$$|\mathbf{v}(t)| = \sqrt{v_x^2(t) + v_y^2(t) + v_z^2(t)}.$$

Speed is not a vector. It is a non-negative number. It tells us how fast the point is moving, but not in which direction.

Velocity tells us both: how fast and in which direction.

3.3.6 Example: motion along a line

Consider the one-dimensional position

$$x(t) = x_0 + at,$$

where x_0 and a are constants.

The velocity is

$$v(t) = \frac{dx}{dt}.$$

Therefore

$$v(t) = a.$$

The velocity is constant. This means that the position changes at the same rate at every time.

If

$$x(t) = 1 + 2t,$$

then

$$v(t) = 2.$$

The number 2 is not only the coefficient in the formula. It is the rate at which position changes with respect to time.

3.3.7 Example: motion along a parabola

Consider

$$\mathbf{r}(t) = (t, t^2).$$

Then

$$x(t) = t,$$

and

$$y(t) = t^2.$$

The velocity is

$$\mathbf{v}(t) = \left(\frac{dx}{dt}, \frac{dy}{dt} \right) = (1, 2t).$$

At $t = 0$,

$$\mathbf{v}(0) = (1, 0).$$

At $t = 1$,

$$\mathbf{v}(1) = (1, 2).$$

At $t = 2$,

$$\mathbf{v}(2) = (1, 4).$$

The trajectory is a parabola, but the velocity tells us how the point moves along the parabola. The horizontal component is constant, while the vertical component changes.

This example also shows that a point on a trajectory and a velocity vector are different objects. The point tells us where the body is. The velocity vector tells us how that position is changing at that moment.

3.3.8 Example: motion along a circle

Consider

$$\mathbf{r}(t) = (R \cos t, R \sin t).$$

The velocity is

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = (-R \sin t, R \cos t).$$

For a concrete point on the circle, take

$$R = 2, \quad t = \frac{\pi}{3}.$$

Then

$$\mathbf{r}\left(\frac{\pi}{3}\right) = \left(2 \cos \frac{\pi}{3}, 2 \sin \frac{\pi}{3}\right) = (1, \sqrt{3}),$$

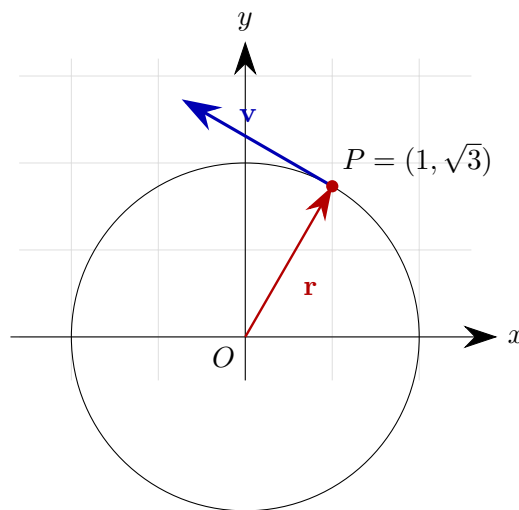
and

$$\mathbf{v}\left(\frac{\pi}{3}\right) = \left(-2 \sin \frac{\pi}{3}, 2 \cos \frac{\pi}{3}\right) = (-\sqrt{3}, 1).$$

Their dot product is

$$(1, \sqrt{3}) \cdot (-\sqrt{3}, 1) = -\sqrt{3} + \sqrt{3} = 0.$$

So at this point the velocity vector is perpendicular to the position vector.



The speed is

$$|\mathbf{v}(t)| = \sqrt{(-R \sin t)^2 + (R \cos t)^2}.$$

Thus

$$|\mathbf{v}(t)| = \sqrt{R^2 \sin^2 t + R^2 \cos^2 t} = R.$$

The speed is constant. However, the velocity vector is not constant, because its direction changes with time.

This is an important point. Constant speed does not mean constant velocity. Velocity is a vector. If the direction changes, the velocity changes.

We can also check that the velocity is tangent to the circle. The position vector is

$$\mathbf{r}(t) = (R \cos t, R \sin t),$$

and the velocity vector is

$$\mathbf{v}(t) = (-R \sin t, R \cos t).$$

Their dot product is

$$\mathbf{r}(t) \cdot \mathbf{v}(t) = R \cos t(-R \sin t) + R \sin t(R \cos t).$$

Therefore

$$\mathbf{r}(t) \cdot \mathbf{v}(t) = 0.$$

The position vector and velocity vector are perpendicular. Since the position vector points radially outward from the center of the circle, the velocity vector is tangent to the circle.

3.3.9 Example: recognizing a special point on a trajectory

Consider a motion in the plane described by

$$\mathbf{r}(t) = (t, -t^2 + 4t).$$

The trajectory is a parabola because

$$x = t,$$

so

$$y = -x^2 + 4x.$$

The velocity is

$$\mathbf{v}(t) = \left(\frac{dx}{dt}, \frac{dy}{dt} \right) = (1, -2t + 4).$$

The vertical component of velocity is

$$v_y(t) = -2t + 4.$$

At the highest point of this parabola, the vertical component of velocity should be zero. We solve

$$-2t + 4 = 0.$$

Thus

$$t = 2.$$

The position at this time is

$$\mathbf{r}(2) = (2, -4 + 8) = (2, 4).$$

So the highest point is $(2, 4)$.

This example shows that velocity is not introduced only to compute another formula. Velocity helps us understand the structure of motion. It can identify special points on a trajectory, such as points where the motion changes from upward to downward.

3.3.10 Units of velocity

Velocity has units of position divided by time. If position is measured in metres and time in seconds, velocity is measured in

$$\text{m/s}.$$

This follows directly from the definition

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}.$$

The numerator has the unit of length, and the denominator has the unit of time.

Whenever a velocity is computed, its unit should be checked. If a calculation gives a velocity in metres or seconds instead of metres per second, something has gone wrong.

3.4 Acceleration

3.4.1 Why velocity is not the end of the description

Velocity describes how position changes. But velocity itself may also change.

A car may move faster and faster. A body may slow down. A point moving along a circle may keep the same speed but continuously change direction. In each of these cases, velocity changes.

Acceleration describes the change of velocity with respect to time.

3.4.2 Average acceleration

Suppose the velocity at time t_1 is $\mathbf{v}(t_1)$, and the velocity at time t_2 is $\mathbf{v}(t_2)$. The change in velocity is

$$\Delta\mathbf{v} = \mathbf{v}(t_2) - \mathbf{v}(t_1).$$

The average acceleration over the interval is

$$\mathbf{a}_{\text{avg}} = \frac{\Delta\mathbf{v}}{\Delta t} = \frac{\mathbf{v}(t_2) - \mathbf{v}(t_1)}{t_2 - t_1}.$$

Average acceleration describes how velocity changes over a time interval.

3.4.3 Instantaneous acceleration

Instantaneous acceleration is the derivative of velocity with respect to time:

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt}.$$

Since velocity is the derivative of position,

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt},$$

we also have

$$\mathbf{a}(t) = \frac{d^2\mathbf{r}}{dt^2}.$$

Thus acceleration is the second derivative of position with respect to time.

In coordinates,

$$\mathbf{a}(t) = \left(\frac{d^2x}{dt^2}, \frac{d^2y}{dt^2}, \frac{d^2z}{dt^2} \right).$$

In two dimensions,

$$\mathbf{a}(t) = \left(\frac{d^2x}{dt^2}, \frac{d^2y}{dt^2} \right).$$

In one dimension,

$$a(t) = \frac{d^2x}{dt^2}.$$

3.4.4 Acceleration is also a vector

Acceleration is a vector. It has magnitude and direction. It describes how the velocity vector changes.

This point is important. Acceleration does not only mean that speed increases. Acceleration means that velocity changes. Since velocity is a vector, it can change in two ways:

- its magnitude can change,
- its direction can change.

Therefore a body may be accelerating even if its speed is constant, provided that the direction of velocity changes.

This is one of the most common conceptual difficulties in kinematics. Constant speed does not necessarily mean zero acceleration.

3.4.5 Example: motion along a line

Consider

$$x(t) = x_0 + at.$$

We already found that

$$v(t) = a.$$

The acceleration is

$$a_{\text{kin}}(t) = \frac{dv}{dt}.$$

Since $v(t)$ is constant,

$$a_{\text{kin}}(t) = 0.$$

Here we use the notation a_{kin} for acceleration in this example to avoid confusion with the parameter a in the formula for $x(t)$. This is a useful warning: the same letter can be used for different things in different contexts, but in one calculation we must keep the meaning clear.

3.4.6 Example: parabolic position

Consider again

$$\mathbf{r}(t) = (t, t^2).$$

We found

$$\mathbf{v}(t) = (1, 2t).$$

Therefore

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = (0, 2).$$

The acceleration is constant. It points in the positive y -direction.

This does not come from looking only at the geometric curve. It comes from how the point moves along the curve as time changes.

3.4.7 Example: circular motion

Consider again

$$\mathbf{r}(t) = (R \cos t, R \sin t).$$

We found

$$\mathbf{v}(t) = (-R \sin t, R \cos t).$$

Now we differentiate again:

$$\mathbf{a}(t) = (-R \cos t, -R \sin t).$$

But

$$\mathbf{r}(t) = (R \cos t, R \sin t).$$

Therefore

$$\mathbf{a}(t) = -\mathbf{r}(t).$$

This means that the acceleration points toward the center of the circle. The velocity is tangent to the circle, but the acceleration points inward.

This is a major conceptual point. In uniform circular motion, speed may be constant, but acceleration is not zero, because the direction of velocity is changing.

3.4.8 Units of acceleration

Acceleration has units of velocity divided by time. If velocity is measured in metres per second and time in seconds, acceleration is measured in

$$\text{m/s}^2.$$

This also follows from

$$\mathbf{a}(t) = \frac{d^2\mathbf{r}}{dt^2}.$$

Position contributes metres, and two differentiations with respect to time contribute two factors of seconds in the denominator.

The unit m/s^2 means metres per second per second. It describes how velocity changes each second.

3.5 Moving between position, velocity, and acceleration

3.5.1 The derivative direction

So far we have moved in one direction:

$$\mathbf{r}(t) \longrightarrow \mathbf{v}(t) \longrightarrow \mathbf{a}(t).$$

This direction uses derivatives:

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt},$$

and

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}.$$

If we know position as a function of time, we can compute velocity and acceleration by differentiating. No additional information is needed for this direction. The function $\mathbf{r}(t)$ already contains all information about the motion.

For example, if

$$\mathbf{r}(t) = (t, t^2),$$

then

$$\mathbf{v}(t) = (1, 2t),$$

and

$$\mathbf{a}(t) = (0, 2).$$

Everything follows by differentiation.

3.5.2 The integral direction

We can also move in the opposite direction. If we know acceleration, we may want to recover velocity. If we know velocity, we may want to recover position. This direction uses integration:

$$\mathbf{a}(t) \longrightarrow \mathbf{v}(t) \longrightarrow \mathbf{r}(t).$$

But integration introduces constants. These constants are not mathematical noise. They represent missing information about the motion.

If

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt},$$

then

$$\mathbf{r}(t) = \int \mathbf{v}(t) dt.$$

More precisely,

$$\mathbf{r}(t) = \mathbf{r}(t_0) + \int_{t_0}^t \mathbf{v}(\tau) d\tau.$$

The symbol τ is used as a dummy variable of integration. The upper limit t is the time at which we want to find the position. The lower limit t_0 is the initial time.

This formula says that the position at time t equals the initial position plus the accumulated effect of velocity from t_0 to t .

Similarly, if

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt},$$

then

$$\mathbf{v}(t) = \mathbf{v}(t_0) + \int_{t_0}^t \mathbf{a}(\tau) d\tau.$$

This formula says that the velocity at time t equals the initial velocity plus the accumulated effect of acceleration from t_0 to t .

3.5.3 Why initial conditions are necessary

Suppose we know only that

$$v(t) = 3.$$

Then

$$x(t) = \int 3 dt = 3t + C.$$

The constant C is not determined by the velocity. It tells us the initial position.

If we know that

$$x(0) = 2,$$

then

$$2 = 3 \cdot 0 + C,$$

so

$$C = 2.$$

Therefore

$$x(t) = 3t + 2.$$

Without the initial condition, we would know the way position changes, but not the actual starting position.

Now suppose we know only that acceleration is constant:

$$a(t) = A.$$

Then velocity is

$$v(t) = At + C_1.$$

The constant C_1 is determined by the initial velocity. If

$$v(0) = v_0,$$

then

$$C_1 = v_0,$$

so

$$v(t) = v_0 + At.$$

To recover position, we integrate again:

$$x(t) = \int (v_0 + At) dt = v_0 t + \frac{1}{2} At^2 + C_2.$$

The constant C_2 is determined by the initial position. If

$$x(0) = x_0,$$

then

$$C_2 = x_0.$$

Therefore

$$x(t) = x_0 + v_0 t + \frac{1}{2} At^2.$$

This formula is often presented as a standard kinematic formula. Here we see where it comes from: it is the result of integrating constant acceleration twice and using initial conditions.

3.5.4 Vector form

The same idea works in vector form.

If acceleration is known, then

$$\mathbf{v}(t) = \mathbf{v}(t_0) + \int_{t_0}^t \mathbf{a}(\tau) d\tau.$$

If velocity is known, then

$$\mathbf{r}(t) = \mathbf{r}(t_0) + \int_{t_0}^t \mathbf{v}(\tau) d\tau.$$

If acceleration is constant and equal to $\mathbf{A}$, then

$$\mathbf{v}(t) = \mathbf{v}_0 + \mathbf{A}t,$$

assuming $t_0 = 0$. Integrating again gives

$$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2} \mathbf{A}t^2.$$

This is the vector version of the same relation.

3.5.5 What is known and what is missing

It is useful to summarize the logic.

If we know position $\mathbf{r}(t)$, then we can find velocity and acceleration by differentiation:

$$\mathbf{r}(t) \longrightarrow \mathbf{v}(t) \longrightarrow \mathbf{a}(t).$$

No initial conditions are needed because the position function already contains the full motion.

If we know velocity $\mathbf{v}(t)$, then we can find acceleration by differentiation. But to find position, we must integrate and we need an initial position:

$$\mathbf{v}(t) \longrightarrow \mathbf{a}(t),$$

but

$$\mathbf{v}(t) \longrightarrow \mathbf{r}(t)$$

requires $\mathbf{r}(t_0)$.

If we know acceleration $\mathbf{a}(t)$, then we can recover velocity and position only by integration. We need initial velocity and initial position:

$$\mathbf{a}(t) \longrightarrow \mathbf{v}(t) \longrightarrow \mathbf{r}(t),$$

but this requires

$$\mathbf{v}(t_0)$$

and

$$\mathbf{r}(t_0).$$

This is not a technical detail. It expresses an important fact about motion: knowing how something changes is not the same as knowing where it is or what its current value is. To reconstruct the motion, we need both a law of change and initial information.

3.5.6 The central structure of kinematics

The central structure of kinematics is therefore:

$$\boxed{\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}}$$

and

$$\boxed{\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}}$$

together with the inverse relations

$$\boxed{\mathbf{v}(t) = \mathbf{v}(t_0) + \int_{t_0}^t \mathbf{a}(\tau) d\tau}$$

and

$$\boxed{\mathbf{r}(t) = \mathbf{r}(t_0) + \int_{t_0}^t \mathbf{v}(\tau) d\tau}.$$

These formulas should not be treated as isolated recipes. They express the whole logic of kinematics:

- position describes where the body is,
- velocity describes how position changes,
- acceleration describes how velocity changes,
- derivatives move from position to velocity and acceleration,
- integrals move back from acceleration or velocity to velocity or position,
- initial conditions supply the information lost when differentiating.

This is the mathematical language that will be used again in later chapters. Dynamics will ask why acceleration appears. But before we ask why motion changes, we must first know how to describe motion and its changes precisely.

Chapter 4

Dynamics

4.1 Introduction

This section provides an introduction to dynamics.

Chapter 5

Oscillations and Waves

5.1 Introduction

This section provides an introduction to oscillations and waves.

Chapter 6

Electromagnetism

6.1 Introduction

This section provides an introduction to electromagnetism.

Chapter 7

Statistics and Measurements

7.1 Introduction

This section provides an introduction to statistics and measurements.

Chapter 8

Cosmology

8.1 Introduction

This section provides an introduction to cosmology.

Chapter 9

Relativity

9.1 Introduction

This section provides an introduction to relativity.

Chapter 10

Modern Physics

10.1 Introduction

This section provides an introduction to modern physics.